

Classifying quadrilaterals



- 1
- a Rectangle
 - b Square
 - c Trapezoid
 - d Rectangle
 - e Trapezoid
- 2
- a Rhombus
 - b Parallelogram
 - c Trapezoid
 - d Rhombus
 - e Trapezoid
 - f Parallelogram
- 3 Parallelogram
- 4
- a False
 - b True
 - c False
 - d False
 - e False
 - f True
 - g False
 - h True
 - i False
 - j True
 - k True
 - l False
 - m False
- 5
- a Rectangle
 - b Parallelogram
 - c Rhombus
 - d Rectangle
 - e Rhombus
 - f Parallelogram
 - g Rhombus
 - h Parallelogram
- 6
- a Parallelogram
 - b Rectangle
- 7
- a There is an additional pair of parallel sides.
 - b Two pairs of opposite sides are congruent.
 - c One pair of opposite sides are parallel.
- 8
- a The parallelogram has four congruent sides.
 - b The parallelogram has four right angles.
 - c The parallelogram has four congruent sides and four right angles.

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- a The trapezoid has diagonals that are congruent.
- b The trapezoid has base angles that are congruent.
- c The trapezoid has opposite angles that are supplementary.

10

To prove: $ABCD$ is a square

Statements	Reasons
1. $ABCD$ is a rectangle and a rhombus	Given
2. $ABCD$ is a parallelogram with four right angles	Definition of a rectangle
3. $ABCD$ is a parallelogram with four congruent sides	Definition of a rhombus
4. $ABCD$ is a square	A parallelogram with four congruent sides and four right angles is a square (definition of a square).

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To prove: $ABCD$ is a rectangle

Statements	Reasons
1. $ABCD$ is a parallelogram and $\overline{AC} \cong \overline{BD}$	Given
2. $\overline{AD} \cong \overline{BC}$	If a quadrilateral is a parallelogram, then it has both pairs of opposite sides congruent.
3. $\overline{AB} \cong \overline{AB}$ and $\overline{CD} \cong \overline{CD}$	Reflexive property of congruence
4. $\triangle ABD \cong \triangle BAC$ and $\triangle ACD \cong \triangle BDC$	Side-side-side congruence theorem
5. $\angle ADC \cong \angle BCD$ and $\angle DAB \cong \angle CBA$	Corresponding parts of congruent triangles are congruent (CPCTC).
6. $\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$	Definition of a parallelogram
7. $\angle ADC$ is supplementary to $\angle BCD$ and $\angle DAB$ is supplementary to $\angle CBA$	If two parallel lines are cut by a transversal, then consecutive interior angles are supplementary (consecutive interior angles theorem).
$m\angle ADC + m\angle BCD = 180^\circ$ and	



Statements		Reasons
9.	$m\angle ADC + m\angle ADC = 180^\circ$ and $m\angle DAB + m\angle DAB = 180^\circ$	Substitution
10.	$2m\angle ADC = 180^\circ$ and $2m\angle DAB = 180^\circ$	Simplify
11.	$m\angle ADC = 90^\circ$ and $m\angle DAB = 90^\circ$	Division property of equality
12.	$\angle ADC, \angle BCD, \angle DAB$ and $\angle CBA$ are right angles	Definition of a right angle
13.	$ABCD$ is a rectangle	Definition of a rectangle

12
a

To prove: $ABCD$ is a rhombus		
Statements		Reasons
1.	$ABCD$ is a parallelogram and $\overline{AB} \cong \overline{BC}$	Given
2.	$\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{DA}$	If a quadrilateral is a parallelogram, then its opposite sides are congruent.
3.	$\overline{AB} \cong \overline{BC} \cong \overline{CD} \cong \overline{DA}$	Transitive property of congruence
4.	$ABCD$ is a rhombus	Definition of a rhombus

b

To prove:  is a rhombus

Statements	Reasons
1. $PQRS$ is a parallelogram and $\overline{PR} \perp \overline{QS}$	Given
2. $\overline{PR}$ and $\overline{QS}$ bisect each other	If a quadrilateral is a parallelogram, then it has two diagonals that bisect one another.
3. $\overline{PT} \cong \overline{TR}$ and $\overline{QT} \cong \overline{TS}$	Definition of a bisector
4. $\angle STP, \angle QTP, \angle QTR, \angle RTS$ are right angles	Definition of perpendicular
5. $\angle STP \cong \angle QTP \cong \angle QTR \cong \angle RTS$	Definition of congruent angles
6. $\triangle STP \cong \triangle QTP \cong \triangle QTR \cong \triangle RTS$	Side-angle-side congruence theorem
7. $\overline{PQ} \cong \overline{QR} \cong \overline{RS} \cong \overline{ST}$	Corresponding parts of congruent triangles are congruent (CPCTC).
8. $PQRS$ is a rhombus	Definition of a rhombus

c


 To prove: $ABCD$ is a rhombus

Statements	Reasons
1. $ABCD$ is a parallelogram and $\overline{BD}$ bisects $\angle ABC$ and $\angle ADC$	Given
2. $\angle 1 \cong \angle 2$ and $\angle 3 \cong \angle 4$	Definition of an angle bisector
3. $\overline{AB} \parallel \overline{CD}$ and $\overline{AD} \parallel \overline{CB}$	Definition of a parallelogram
4. $\angle 1 \cong \angle 3$ and $\angle 2 \cong \angle 4$	If two lines are parallel, then alternate interior angles are congruent (alternate interior angles theorem).
5. $\angle 1 \cong \angle 4$ and $\angle 2 \cong \angle 3$	Transitive property of congruence
6. $\overline{DA} \cong \overline{AB}$ and $\overline{BC} \cong \overline{CD}$	If two angles of a triangle are congruent, then the sides opposite those angles are congruent (base angles converse).
7. $\overline{AB} \cong \overline{DC}$ and $\overline{AD} \cong \overline{BC}$	If a quadrilateral is a parallelogram, then the opposite sides are congruent.
8. $\overline{AB} \cong \overline{BC} \cong \overline{CD} \cong \overline{DA}$	Transitive property of congruence
9. $ABCD$ is a rhombus	Definition of a rhombus

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a

To prove: $ABCD$ is an isosceles trapezoid

Statements	Reasons
1. $ABCD$ is a trapezoid, $\angle ADC \cong \angle BCD$, and $\angle APD$ and $\angle BQC$ are right angles	Given
2. $\angle APD \cong \angle BQC$	All right angles are congruent.
3. $\overline{AP} \cong \overline{BQ}$	Altitudes of a trapezoid are congruent
4. $\triangle APD \cong \triangle BQC$	Angle-angle-side congruence theorem
5. $\overline{AD} \cong \overline{BC}$	Corresponding parts of congruent triangles are congruent (CPCTC)
6. $ABCD$ is an isosceles trapezoid	Definition of an isosceles trapezoid

b

To prove: $ABCD$ isosceles trapezoid



Statements	Reasons
1. $ABCD$ is a trapezoid, $\angle A$ and $\angle C$ are supplementary	Given
2. $\overline{AB} \parallel \overline{DC}$	Definition of a trapezoid
3. $\angle A$ and $\angle D$ are supplementary	If two lines are parallel, then consecutive interior angles are supplementary (consecutive interior angles postulate).
4. $\angle C \cong \angle D$	Supplements of the same angle or congruent angles are congruent (congruent supplements theorem).
5. $ABCD$ is an isosceles trapezoid	If a trapezoid has base angles congruent, then it is an isosceles trapezoid.

c

To prove: $ABCD$ isosceles trapezoid

Statements	Reasons
1. $ABCD$ is a trapezoid, $\overline{AC} \cong \overline{BD}$ and $\overline{AC} \parallel \overline{BP}$	Given
2. $\overline{AB} \parallel \overline{DC}$	Definition of a trapezoid
3. $ABPC$ is a parallelogram	Definition of a parallelogram
4. $\overline{AC} \cong \overline{BP}$	If a quadrilateral is a parallelogram, then its opposite sides are congruent.
5. $\overline{BD} \cong \overline{BP}$	Transitive property of congruence
6. $\angle BDC \cong \angle BPC$	If two sides of a triangle are congruent, then the angles opposite those sides are congruent (base angles theorem).
7. $\angle ACD \cong \angle BPC$	If two lines are parallel, then corresponding angles are congruent (corresponding angles theorem).
8. $\angle BDC \cong \angle ACD$	Transitive property of congruence
9. $\overline{DC} \cong \overline{DC}$	Reflexive property of congruence
10. $\triangle ACD \cong \triangle BDC$	Side-angle-side congruence theorem (SAS)
11. $\overline{AD} \cong \overline{BC}$	Corresponding parts of congruent triangles are congruent (CPCTC)
12. $ABCD$ is an isosceles trapezoid	Definition of an isosceles trapezoid

Additional questions

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- | | |
|--------------------------------|---|
| a Kite | b Rectangle, rhombus, parallelogram |
| c Rhombus, kite, parallelogram | d Parallelogram |
| e Kite | f Rectangle, rhombus, kite, parallelogram |
| g Kite | h Rectangle, rhombus, parallelogram |
| i Parallelogram | j Trapezoid |
| k Neither | l Parallelogram |



15 3, because otherwise the angle sum would be larger than 360.

16 4, this is the maximum number of angles a quadrilateral can have, and it would give the correct angle sum. Squares and rectangles have four right-angles.

17 Yes, a trapezium.

18 No